

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**An Intelligent Machine Learning Driven Framework for
Predictive Risk Stratification of Diabetes Mellitus**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Muskan

2210990588

Supervised By:

Dr. Isha Kansal



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

CONTENTS

S.No.	Title	Page No.
	Acknowledgement	4
	List of Figures and Tables	5
	Abstract	6
1	Introduction	7-8
2	Methodology	9-12
3	Tools and Technologies	13-15
4	Implementation	16-22
5	Results	23-24
6	Conclusion and Future Scope	25
	References	26

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**An Intelligent Machine Learning Driven Framework for Predictive Risk Stratification of Diabetes Mellitus**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Isha Kansal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Muskan

Date: 19th May 2026

(University Roll No - 2210990588)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Isha Kansal

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide for their continuous support, valuable guidance, and encouragement throughout the development of this project. Their insightful suggestions and constant motivation helped me in successfully completing this work.

I am also thankful to the faculty members of the Computer Science Department for providing the necessary resources and academic support required for this project. Their guidance played an important role in enhancing my knowledge and understanding.

I would like to extend my gratitude to my institution for providing a conducive environment and facilities to carry out this project successfully.

Finally, I would like to thank my friends and family for their constant support, encouragement, and cooperation throughout the project.

LIST OF FIGURES AND TABLES

Fig 1	Machine Learning Workflow
Fig 2	Machine Learning Pipeline for Healthcare Data
Fig3	Supervised Learning Process
Fig 4	Implementation flowchart
Fig 5	ROC Curve Comparison
Fig 6	SHAP summary (beeswarm) plot showing feature impact on XGBoost predictions
Fig 7	SHAP Waterfall Plot Showing Feature Contributions in XGBoost
Fig 8	LIME-based local explanation for XGBoost prediction
Fig 9	Permutation feature importance for the XGBoost model.
Fig 10	Partial Dependency plot for the XGBoost model.
Fig 11	Global feature importance of the XGBoost model based on mean absolute SHAP values
Fig 12	Model Comparison
Table	Results

ABSTRACT

Diabetes Mellitus is a rapidly growing global health concern characterized by elevated blood glucose levels, leading to severe complications such as cardiovascular diseases, kidney failure, and nerve damage if not detected at an early stage. With the increasing availability of healthcare data, machine learning techniques provide an efficient and scalable approach for early prediction and risk stratification of diabetes.

This project proposes an intelligent machine learning-driven framework for predictive risk stratification of Diabetes Mellitus using multiple supervised learning algorithms. The framework utilizes a structured dataset containing key medical attributes such as glucose level, blood pressure, BMI, insulin levels, and age. Data preprocessing techniques, including handling missing values and feature scaling, are applied to ensure data quality and consistency.

A comparative analysis of various machine learning models—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost)—is performed to identify the most effective predictive model. The models are evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. To enhance model interpretability and trustworthiness, Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are integrated. These techniques provide both global and local explanations, enabling a better understanding of how individual features influence predictions.

The results demonstrate that ensemble learning methods, particularly XGBoost, outperform other models in predictive accuracy and robustness. The proposed framework not only achieves high performance but also ensures transparency, making it suitable for real-world healthcare applications where interpretability is critical.

1. INTRODUCTION

1.1 Background

Diabetes Mellitus is a chronic metabolic disorder that has become a major global health concern due to its increasing prevalence and long-term complications. It occurs when the body is unable to effectively regulate blood glucose levels due to insufficient insulin production or resistance to insulin. If left undiagnosed or untreated, diabetes can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, and vision loss.

With the rapid growth of digital healthcare systems and medical data, traditional diagnostic methods are increasingly being complemented by intelligent computational approaches. Machine learning has emerged as a powerful tool in the healthcare domain, enabling the analysis of complex datasets and supporting early disease prediction. This project focuses on leveraging machine learning techniques to develop a predictive system for diabetes risk stratification.

1.2 Problem Statement

Early detection of diabetes remains a challenge due to the lack of efficient predictive tools that can analyze multiple medical parameters simultaneously. Traditional methods rely heavily on manual diagnosis and clinical expertise, which may lead to delayed detection and increased risk of complications. There is a need for an automated, accurate, and interpretable system that can assist in identifying individuals at risk of diabetes at an early stage.

1.3 Objectives

1. To develop a machine learning-based predictive model for diabetes risk classification.
2. To compare the performance of multiple classification algorithms.
3. To evaluate models using standard performance metrics.
4. To integrate Explainable AI techniques for model transparency.
5. To identify key factors influencing diabetes prediction.

1.4 Scope

The scope of this project focuses on the development and analysis of a machine learning-based system for predicting the risk of Diabetes Mellitus using a structured healthcare dataset. The system utilizes supervised machine learning algorithms to classify individuals as diabetic or non-diabetic based on various medical attributes such as glucose level, BMI, age, insulin, and blood pressure.

A key aspect of this project is the integration of Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME, which provide insights into model predictions. These techniques help in understanding feature importance and ensure transparency, making the system suitable for healthcare decision support applications.

The project also includes performance evaluation using various metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, along with visualization techniques like bar charts and ROC curves for better interpretation of results. Overall, this project provides a strong foundation for machine learning-based healthcare prediction systems and can be further extended for real-world applications in future work.

2. METHODOLOGY

The proposed methodology follows a structured pipeline consisting of data preprocessing, model training, evaluation, and interpretability analysis.

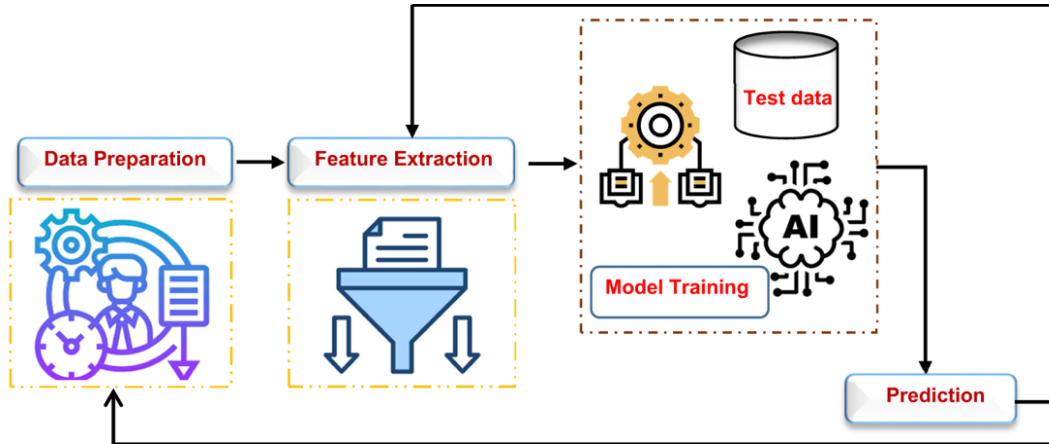


Figure 1 Machine Learning Workflow

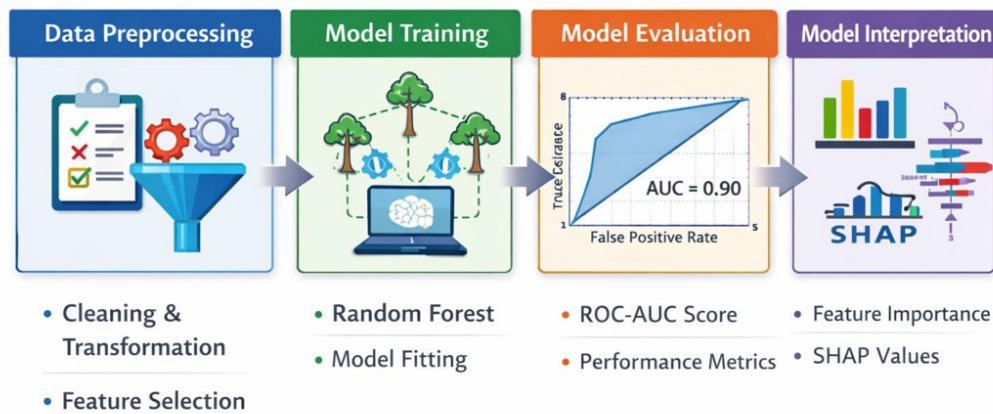


Figure 2 Machine Learning Pipeline for Healthcare Data

1. Data Collection

The dataset used in this project is a publicly available diabetes dataset containing medical records of patients. It includes features such as:

- Pregnancies
- Glucose Level
- Blood Pressure

- Skin Thickness
- Insulin Level
- Body Mass Index (BMI)
- Diabetes Pedigree Function
- Age

The target variable indicates whether a patient is diabetic (1) or non-diabetic (0).

2. Data Preprocessing

Data preprocessing is a critical step to ensure the quality and reliability of the model.

- **Handling Missing Values:**

Certain features such as Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI contain zero values, which are not realistic. These values are replaced with NaN and then imputed using median values.

- **Feature Scaling:**

Standardization is applied using Standard Scaler to normalize feature values and improve model performance, especially for algorithms like SVM and Logistic Regression.

- **Train-Test Split:**

The dataset is divided into training (80%) and testing (20%) sets using stratified sampling to maintain class balance.

3. Model Development

Multiple Supervised Machine Learning models are implemented to compare performance:

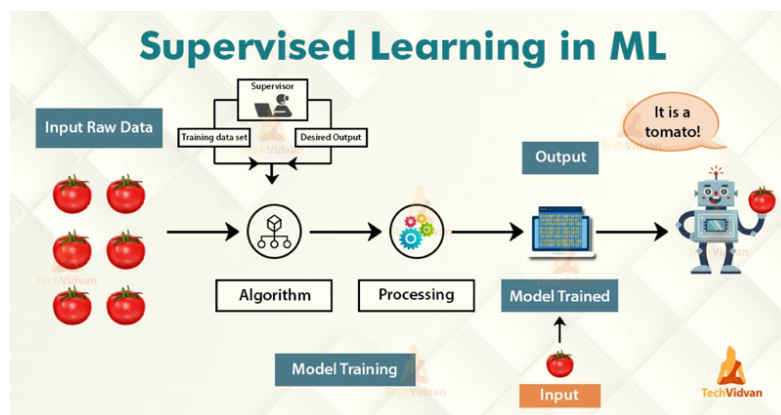


Figure 3 Supervised Learning Process

- **Logistic Regression:** A baseline linear model for classification
- **Decision Tree:** A non-linear model that splits data based on feature values
- **Random Forest:** An ensemble of decision trees for improved accuracy
- **Support Vector Machine (SVM):** A model that finds optimal hyperplanes for classification
- **XGBoost:** An advanced boosting algorithm known for high performance and efficiency

Each model is trained using the scaled training dataset.

4. Model Evaluation

The performance of each model is evaluated using multiple metrics:

- **Accuracy:** Overall correctness of the model
- **Precision:** Correct positive predictions
- **Recall:** Ability to identify actual positives
- **F1 Score:** Balance between precision and recall
- **ROC-AUC:** Measures model's ability to distinguish between classes

Visualizations such as bar charts and ROC curves are used for comparison.

5. Explainable AI (XAI) Integration

To ensure interpretability, several XAI techniques are applied:

- **SHAP (SHapley Additive Explanations):**
Provides global and local explanations by assigning importance values to each feature.
- **LIME (Local Interpretable Model-Agnostic Explanations):**
Explains individual predictions by approximating the model locally.
- **Permutation Importance:**
Measures feature importance by evaluating the effect of feature shuffling.
- **Partial Dependence Plots (PDP):**
Show the relationship between selected features and model predictions.

6. Visualization and Analysis

Graphical representations are used to interpret results:

- Bar plots for model comparison
- ROC curve visualization
- Feature importance graphs
- SHAP summary plots

7. Final Model Selection

Based on evaluation metrics and interpretability, the best-performing model (XGBoost) is selected as the final model for diabetes risk prediction.

3. TOOLS AND TECHNOLOGIES

The development of the proposed machine learning framework for diabetes risk prediction involves a combination of programming languages, libraries, and computational platforms. These tools enable efficient data processing, model development, evaluation, and visualization.

1. Programming Language

- **Python**

Python is used as the primary programming language due to its simplicity, extensive library support, and strong community in data science and machine learning. It allows rapid development and easy integration of various ML and visualization tools.

2. Libraries and Frameworks

a) Data Manipulation and Analysis

- **Pandas**

Used for handling structured data in tabular form (DataFrames). It is utilized for data loading, cleaning, preprocessing, and transformation.

- **NumPy**

Provides support for numerical operations and array manipulations. It is used for efficient computation and handling missing values.

b) Machine Learning and Preprocessing

- **Scikit-learn (sklearn)**

A comprehensive library used for:

- Data preprocessing (StandardScaler)
- Model building (Logistic Regression, Decision Tree, Random Forest, SVM)
- Model evaluation (Accuracy, Precision, Recall, F1 Score, ROC-AUC)
- Data splitting (train_test_split)

c) Advanced Machine Learning

- **XGBoost (Extreme Gradient Boosting)**

An optimized gradient boosting algorithm known for high performance and scalability. It is used to build a robust predictive model that outperforms traditional algorithms.

d) Data Visualization

- **Matplotlib**

Used to create graphical representations such as:

- Bar charts for model comparison
- ROC curves
- Feature importance plots

These visualizations help in understanding model performance and results clearly.

e) Explainable Artificial Intelligence (XAI)

- **SHAP (SHapley Additive Explanations)**

Used to interpret model predictions by assigning importance values to each feature. It provides:

- Global explanations (overall feature importance)
- Local explanations (individual predictions)

- **LIME (Local Interpretable Model-Agnostic Explanations)**

Explains predictions of any model locally by approximating it with an interpretable model. It helps understand why a specific prediction was made.

f) Model Interpretation Tools

- **Permutation Importance (sklearn)**

Measures feature importance by randomly shuffling feature values and observing the impact on model performance.

- **Partial Dependence Plot (PDP)**

Shows the relationship between selected features and predicted outcomes, helping in understanding feature influence.

3. Development Environment

- **Google Colab / Jupyter Notebook**

Used as the development platform for:

- Writing and executing Python code
- Visualizing outputs and graphs

- Running experiments interactively

These environments provide GPU support and easy integration with libraries.

4. Dataset

- A publicly available diabetes dataset (CSV format) is used.
- It contains medical attributes such as glucose level, BMI, insulin, and age.
- The dataset is used for training and testing machine learning models.

4. IMPLEMENTATION

4.1 Implementation Details

The implementation of the proposed intelligent machine learning framework follows a systematic step-by-step pipeline, beginning with data loading and progressing through preprocessing, model development, evaluation, visualization, and interpretability. Each stage is carefully designed to ensure accuracy, efficiency, and reliability of the predictive system.

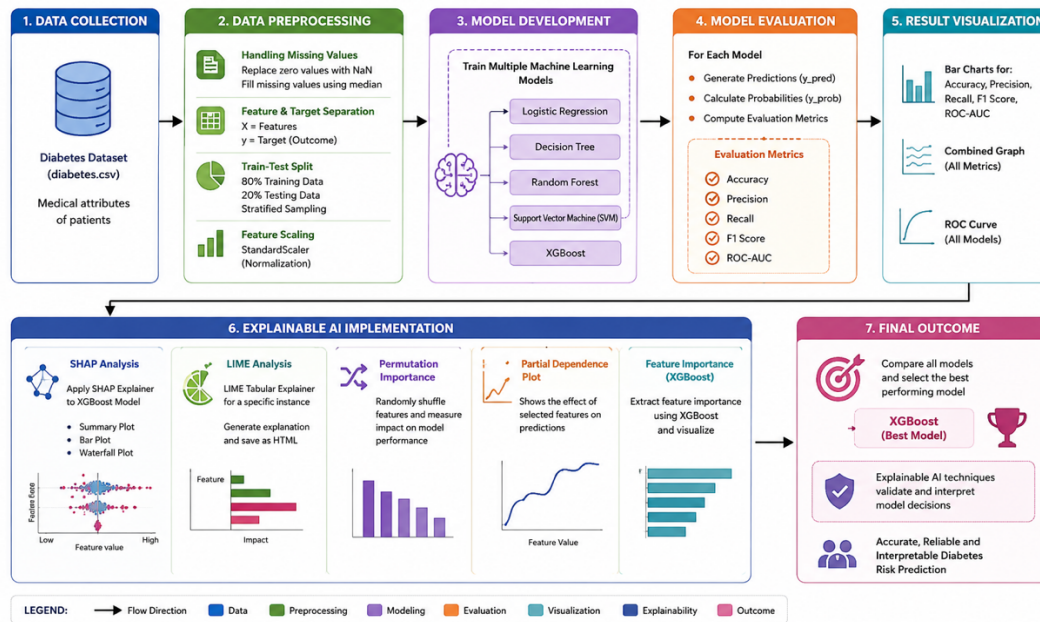


Figure 4 Implementation flowchart

1. Library Installation and Import

The implementation process begins with the installation and import of all necessary libraries required for data processing, machine learning, visualization, and model interpretability. These libraries provide essential functionalities for building and evaluating the predictive system. Pandas and NumPy are used for efficient data handling and numerical computations. Pandas enables structured data manipulation using DataFrames, while NumPy supports array-based operations and mathematical functions. Scikit-learn is used extensively for model development, preprocessing, and evaluation. It provides built-in implementations of various machine learning algorithms as well as tools for scaling, splitting, and performance measurement.

XGBoost is included as an advanced gradient boosting algorithm that enhances predictive accuracy and performance. For model interpretability, SHAP and LIME are integrated into the system, allowing both global and local explanations of predictions.

2. Data Loading

The dataset is loaded using the Pandas library by reading the CSV file named *diabetes.csv*. The data is stored in a DataFrame, which allows easy access, manipulation, and analysis of the dataset. The dataset consists of multiple medical features such as glucose level, blood pressure, BMI, insulin level, and age, along with a target variable labeled *Outcome*, which indicates whether a patient is diabetic or not.

3. Data Preprocessing

Data preprocessing is a crucial step to ensure that the dataset is clean, consistent, and suitable for model training.

a) Handling Missing Values

Certain columns such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI contain zero values, which are not realistic in a medical context. These values are treated as missing and replaced with NaN values. To handle these missing values, median imputation is applied, where each missing value is replaced with the median of the respective feature. This approach helps maintain the distribution of the data and reduces the impact of outliers.

b) Feature and Target Separation

The dataset is divided into two components: input features and target variable. The independent variables (features) are stored in matrix X , while the dependent variable (target), which represents the diabetes outcome, is stored in vector y .

c) Train-Test Split

The dataset is split into training and testing sets using an 80:20 ratio. This ensures that the majority of the data is used for training while a portion is reserved for evaluating model performance. Stratified sampling is applied to maintain a balanced distribution of diabetic and non-diabetic cases in both sets.

d) Feature Scaling

Feature scaling is performed using StandardScaler to normalize the data. This ensures that all features have a mean of zero and a standard deviation of one, allowing models to perform more efficiently. Scaling is particularly important for algorithms such as Support Vector Machine and Logistic Regression, which are sensitive to feature magnitudes.

4. Model Development

In this stage, multiple machine learning models are implemented to perform classification and enable comparative analysis. A dictionary of models is created, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.

Each model is trained using the scaled training dataset. After training, the models are used to make predictions on the test dataset. This multi-model approach allows for a comprehensive comparison of performance and helps in selecting the most suitable algorithm for diabetes prediction.

5. Model Evaluation

The performance of each model is evaluated using several standard classification metrics. For each model, predicted labels (y_pred) and probability scores (y_prob) are generated.

The following evaluation metrics are computed:

1. Accuracy measures the overall correctness of the model.
2. Precision evaluates how many of the predicted positive cases are actually correct.
3. Recall measures the model's ability to correctly identify actual positive cases.
4. F1 Score provides a balance between precision and recall.
5. ROC-AUC measures the model's ability to distinguish between classes.

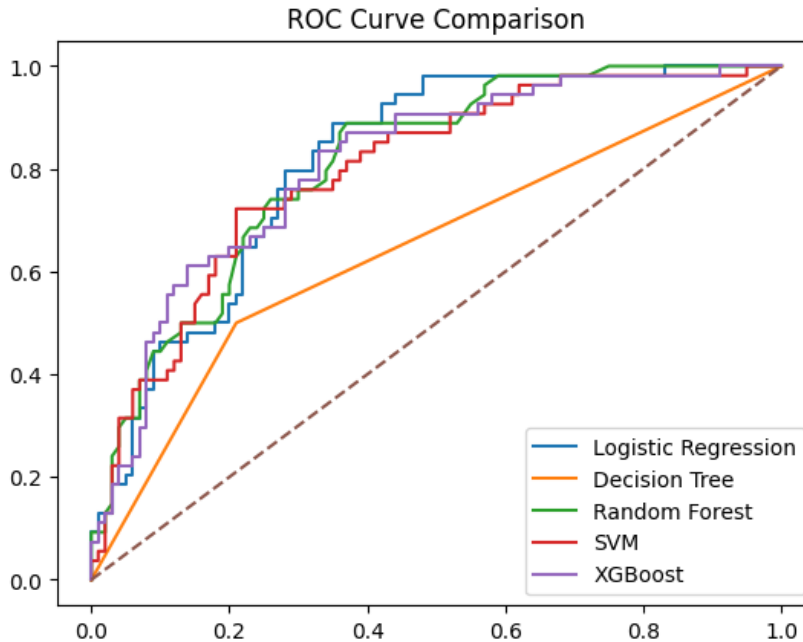


Figure 5 ROC Curve Comparison

All evaluation results are stored in a structured DataFrame, allowing easy comparison across models.

6. Result Visualization

Visualization techniques are used to analyze and compare the performance of different models effectively. Various graphs are plotted using Matplotlib to provide clear insights into model behavior. Bar charts are created to compare different performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC across all models. A combined graph is also generated to display multiple metrics in a single visualization, making comparison easier.

Additionally, ROC curves are plotted for all models. These curves illustrate the trade-off between true positive rate and false positive rate, providing a visual representation of classification performance.

7. Explainable AI Implementation

To enhance model transparency and interpretability, Explainable AI techniques are integrated into the system.

a) SHAP Analysis

SHAP Explainer is applied to the XGBoost model to compute feature importance values. It generates summary plots that highlight the overall importance of features, bar plots for global importance, and waterfall plots for explaining individual predictions. This helps in understanding how each feature contributes to the model's output.

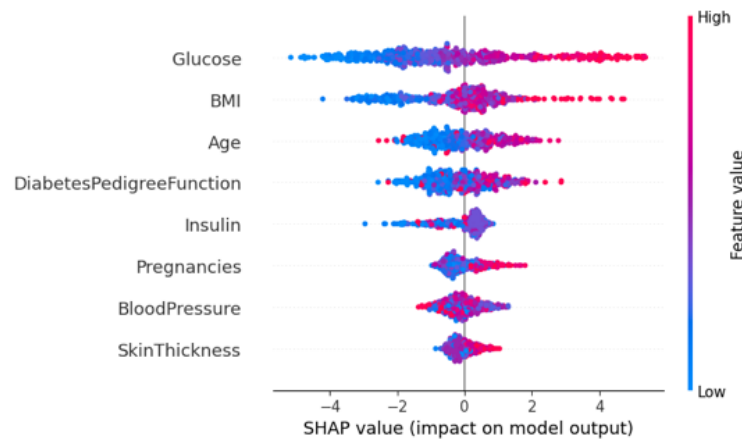


Figure 6 SHAP summary (beeswarm) plot showing feature impact on XGBoost predictions

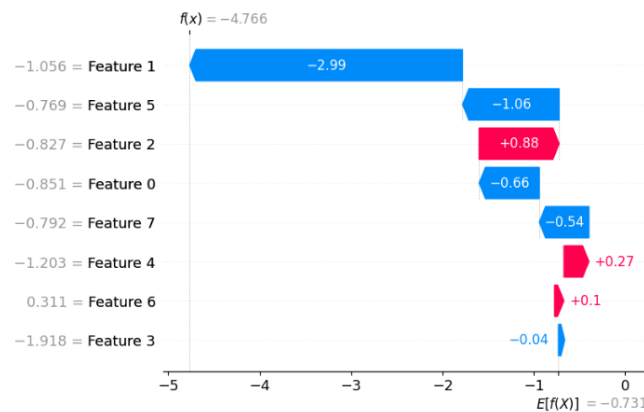


Figure 7 SHAP Waterfall Plot Showing Feature Contributions in XGBoost

b) LIME Analysis

LIME Tabular Explainer is initialized using the training dataset. A specific instance from the test dataset is selected, and LIME generates a local explanation for that prediction. The explanation is saved as an HTML file for visualization and further analysis.

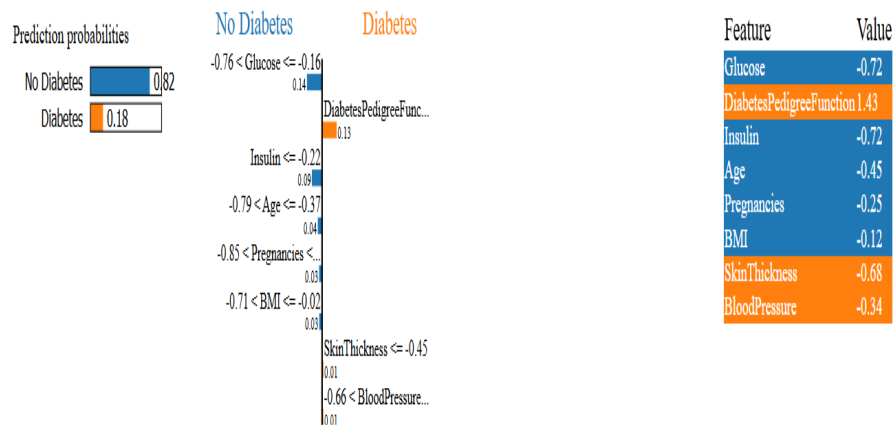


Figure 8 LIME-based local explanation for XGBoost prediction

c) Permutation Importance

Permutation importance is used to measure the significance of each feature by randomly shuffling its values and observing the change in model performance. This method provides an additional validation of feature importance.

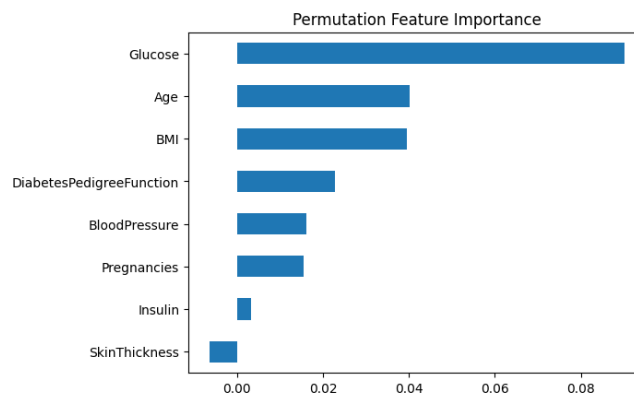


Figure 9 Permutation feature importance for the XGBoost model.

d) Partial Dependence Plot

Partial dependence plots are generated to show how selected features influence model predictions. These plots help in understanding the relationship between feature values and predicted outcomes.

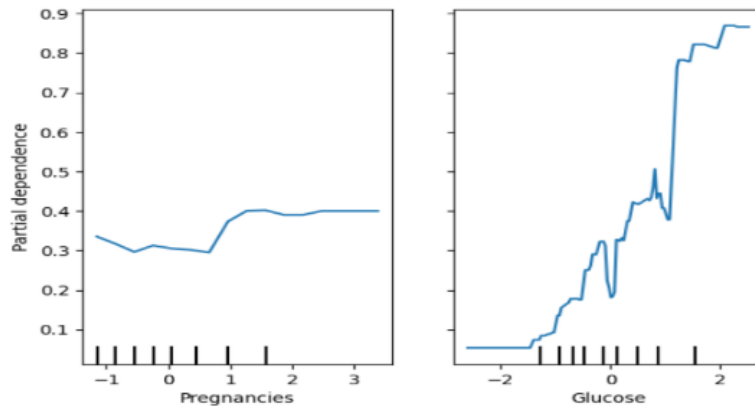


Figure 10 Partial Dependency plot for the XGBoost model.

e) Feature Importance (XGBoost)

Feature importance scores are extracted directly from the XGBoost model and visualized using bar charts. This provides a clear ranking of features based on their contribution to the prediction process.

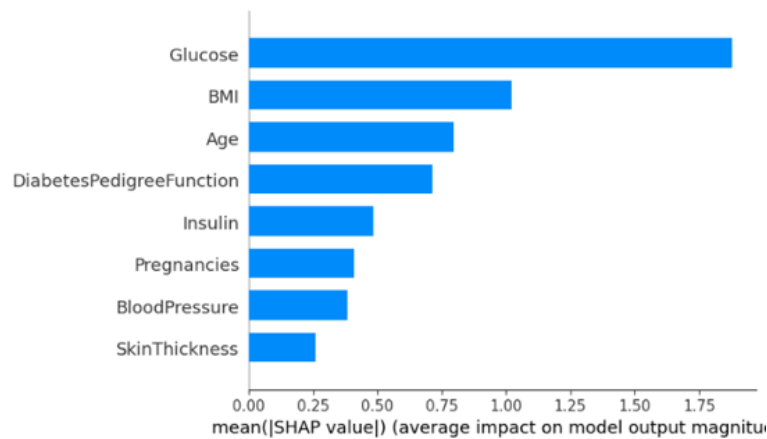


Figure 11 Global feature importance of the XGBoost model based on mean absolute SHAP values

6. RESULTS

6.1 Model Performance and Analysis

Model	Accuracy	Precision	Recall	F1score	ROC-AUC
Random Forest	0.753	0.682	0.556	0.612	0.812
XGBoost	0.760	0.673	0.611	0.641	0.808
SVM	0.740	0.652	0.556	0.612	0.812
Decision Tree	0.682	0.551	0.500	0.524	0.640
Logistic Regression	0.708	0.600	0.500	0.545	0.813

The results obtained from the implementation clearly demonstrate that machine learning models can effectively predict the risk of Diabetes Mellitus using medical attributes. Multiple classification algorithms were trained and evaluated, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.

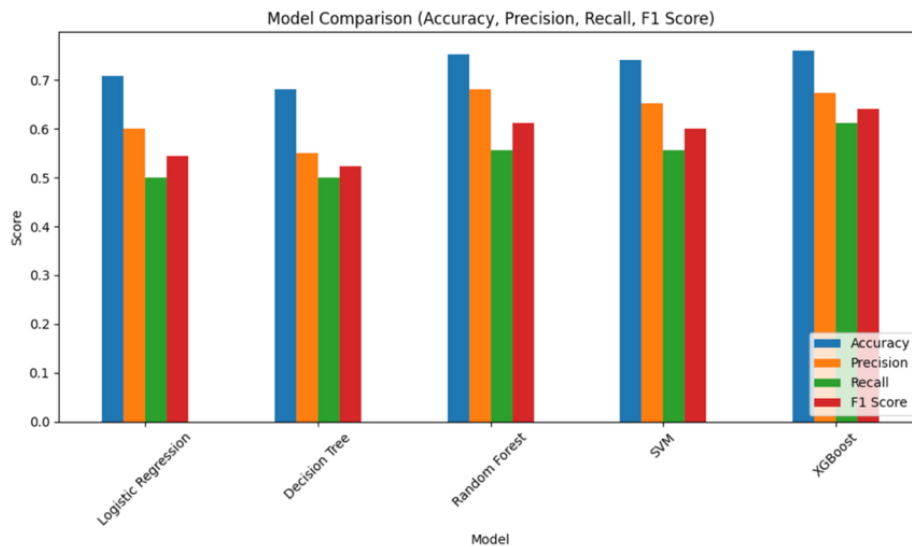


Figure 12 Model Comparison

Among all the models, XGBoost achieved the best performance in terms of accuracy and ROC-AUC score. This is mainly due to its ability to handle complex patterns and combine multiple weak learners into a strong predictive model. Random Forest also performed well due to its ensemble nature, while Logistic Regression provided stable baseline results. SVM showed good performance after feature scaling but was comparatively less effective than ensemble models. The evaluation metrics used in the analysis provided a comprehensive understanding of model performance:

1. Accuracy measured overall correctness of predictions.
2. Precision indicated how many predicted diabetic cases were actually correct.
3. Recall highlighted the model's ability to detect actual diabetic patients, which is critical in healthcare.
4. F1 Score balanced precision and recall.
5. ROC-AUC measured the model's ability to distinguish between classes.

6.2 Feature Insights and Explainability

Explainable AI techniques played a crucial role in understanding the behavior of the model. SHAP analysis revealed that features such as glucose level, BMI, age, and insulin level are the most significant contributors to diabetes prediction. These features have a strong influence on the final outcome and are medically relevant. LIME was used to explain individual predictions, helping to understand why a particular patient was classified as diabetic or non-diabetic. This local interpretability makes the system more transparent and trustworthy. Additional techniques such as permutation importance and partial dependence plots further validated the importance of key features and showed how changes in feature values impact predictions.

6.3 Overall Outcome

The final outcome of the project is a reliable and interpretable machine learning framework for diabetes risk prediction. The system successfully compares multiple models and identifies the best-performing one. The integration of Explainable AI ensures that the predictions are not only accurate but also understandable, making the system suitable for real-world healthcare applications.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project successfully developed an intelligent machine learning-based framework for predicting the risk of Diabetes Mellitus. Multiple machine learning models were implemented and evaluated to determine the most effective approach for classification.

The results indicate that XGBoost outperforms other models in terms of accuracy and ROC-AUC score, making it the most suitable model for this task. The integration of Explainable AI techniques such as SHAP and LIME enhances the transparency of the system by providing insights into feature importance and individual predictions.

Overall, the framework demonstrates the potential of machine learning in early disease detection and risk assessment. It can assist healthcare professionals in making informed decisions and improve patient outcomes through timely diagnosis.

7.2 Future Scope

The project can be extended and improved in several ways to increase its practical applicability and performance:

1. Integration with real-time hospital databases for live prediction.
2. Development of a web or mobile application for easy access.
3. Inclusion of additional data such as lifestyle, diet, and genetic factors.
4. Use of deep learning models for improved accuracy on large datasets.
5. Deployment on cloud platforms for scalability and real-time monitoring.
6. Continuous model updates using new data to improve performance over time.

8. REFERENCES

The following references were used during the development of this project:

<https://www.geeksforgeeks.org/machine-learning/machine-learning/>

https://scikit-learn.org/stable/user_guide.html

https://xgboost.readthedocs.io/en/release_3.2.0/

<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>

<https://www.sciencedirect.com/science/article/pii/S1877050920300557>

<https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2018.00515/full>